

1) Use Trapezoidal Rule to approximate the definite integral $\int_3^5 \sqrt{x^2 - 4} \cdot dx$

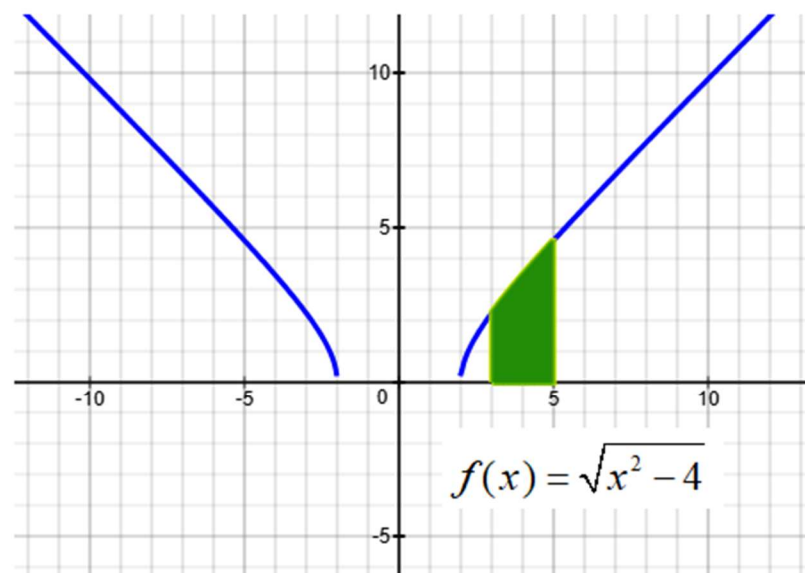
Let $n = 10$; $a = 3$; $b = 5$; $f(x) = \sqrt{x^2 - 4}$

$$\int_3^5 \sqrt{x^2 - 4} \cdot dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \dots + 2f(x_9) + f(x_{10})]$$

$$\Delta x = \frac{b-a}{n} = \frac{2}{10} = 0.2.$$

Complete the following table:

x_i	$f(x_i)$
$x_0 = a = 3$	$f(x_0) = \underline{\quad ? \quad}$
$x_1 = 3 + 1 \cdot \Delta x = 3.2$	$f(x_1) = \underline{\quad ? \quad}$
$x_2 = 3 + 2 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_2) = \underline{\quad ? \quad}$
$x_3 = 3 + 3 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_3) = \underline{\quad ? \quad}$
$x_4 = 3 + 4 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_4) = \underline{\quad ? \quad}$
$x_5 = 3 + 5 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_5) = \underline{\quad ? \quad}$
$x_6 = 3 + 6 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_6) = \underline{\quad ? \quad}$
$x_7 = 3 + 7 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_7) = \underline{\quad ? \quad}$
$x_8 = 3 + 8 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_8) = \underline{\quad ? \quad}$
$x_9 = 3 + 9 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_9) = \underline{\quad ? \quad}$
$x_{10} = 3 + 10 \cdot \Delta x = 5$	$f(x_{10}) = \underline{\quad ? \quad}$



$$\int_3^5 \sqrt{x^2 - 4} \cdot dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \dots + 2f(x_9) + f(x_{10})] = \underline{\quad ? \quad}$$

2) Use Trapezoidal Rule to approximate the definite integral $\int_0^3 \sqrt{x^3 + 1} \cdot dx$

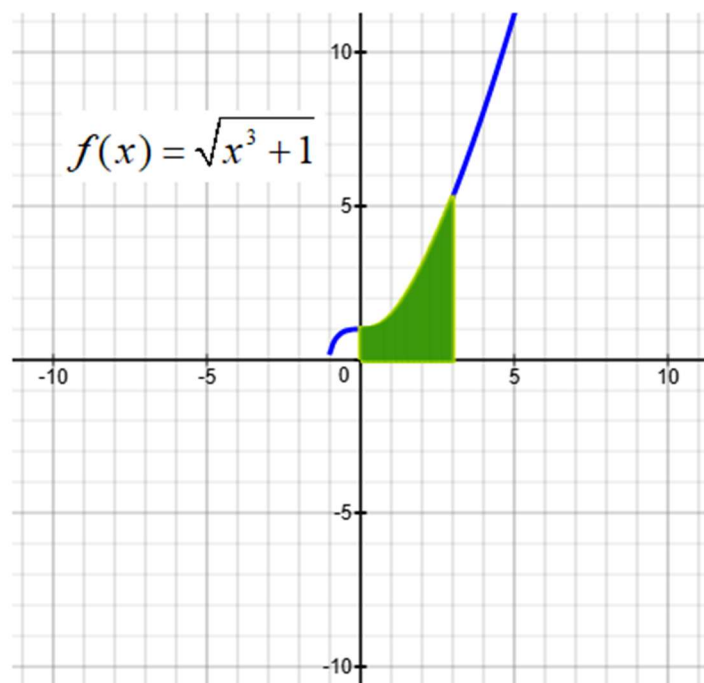
Let $n = 10$; $a = 0$; $b = 3$; $f(x) = \sqrt{x^3 + 1}$

$$\int_0^3 \sqrt{x^3 + 1} \cdot dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \dots + 2f(x_9) + f(x_{10})]$$

$$\Delta x = \frac{b-a}{n} = \frac{3}{10} = 0.3.$$

Complete the following table:

x_i	$f(x_i)$
$x_0 = 0$	$f(x_0) = \underline{\quad ? \quad}$
$x_1 = 1 \cdot \Delta x = 0.3$	$f(x_1) = \underline{\quad ? \quad}$
$x_2 = 2 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_2) = \underline{\quad ? \quad}$
$x_3 = 3 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_3) = \underline{\quad ? \quad}$
$x_4 = 4 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_4) = \underline{\quad ? \quad}$
$x_5 = 5 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_5) = \underline{\quad ? \quad}$
$x_6 = 6 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_6) = \underline{\quad ? \quad}$
$x_7 = 7 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_7) = \underline{\quad ? \quad}$
$x_8 = 8 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_8) = \underline{\quad ? \quad}$
$x_9 = 9 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_9) = \underline{\quad ? \quad}$
$x_{10} = 10 \cdot \Delta x = 3$	$f(x_{10}) = \underline{\quad ? \quad}$



$$\int_0^3 \sqrt{x^3 + 1} \cdot dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \dots + 2f(x_9) + f(x_{10})] = \underline{\quad ? \quad}$$

3) Use Trapezoidal Rule to approximate the definite integral $\int_0^2 x \cdot \sqrt{x^4 + 1} \cdot dx$

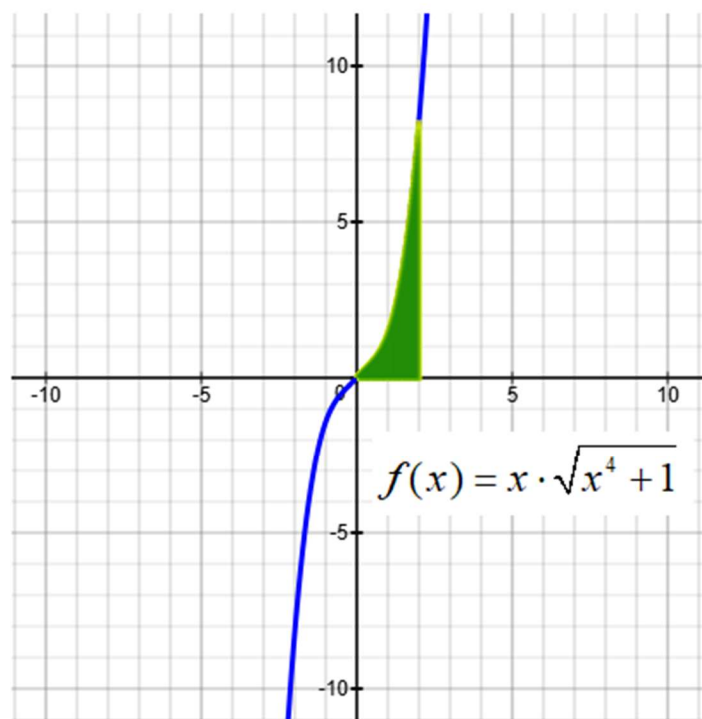
Let $n = 10$; $a = 0$; $b = 2$; $f(x) = x \cdot \sqrt{x^4 + 1}$

$$\int_0^2 x \cdot \sqrt{x^4 + 1} \cdot dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \dots + 2f(x_9) + f(x_{10})]$$

$$\Delta x = \frac{b-a}{n} = \frac{2}{10} = 0.2.$$

Complete the following table:

x_i	$f(x_i)$
$x_0 = 0$	$f(x_0) = \underline{\quad ? \quad}$
$x_1 = 1 \cdot \Delta x = 0.2$	$f(x_1) = \underline{\quad ? \quad}$
$x_2 = 2 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_2) = \underline{\quad ? \quad}$
$x_3 = 3 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_3) = \underline{\quad ? \quad}$
$x_4 = 4 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_4) = \underline{\quad ? \quad}$
$x_5 = 5 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_5) = \underline{\quad ? \quad}$
$x_6 = 6 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_6) = \underline{\quad ? \quad}$
$x_7 = 7 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_7) = \underline{\quad ? \quad}$
$x_8 = 8 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_8) = \underline{\quad ? \quad}$
$x_9 = 9 \cdot \Delta x = \underline{\quad ? \quad}$	$f(x_9) = \underline{\quad ? \quad}$
$x_{10} = 10 \cdot \Delta x = 2$	$f(x_{10}) = \underline{\quad ? \quad}$



$$\int_0^2 x \cdot \sqrt{x^4 + 1} \cdot dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \dots + 2f(x_9) + f(x_{10})] = \underline{\quad ? \quad}$$

